

Microscopic.Game App Summary

WHAT IT IS

Browser-based microscopic organism game prototype with a launcher, live tuning drawer, and canvas simulation. It can start randomized sessions or load JSON-defined levels, and the repo also includes a separate in-browser level editor that exports playable level JSON.

WHO IT'S FOR

Primary persona: a game designer-developer iterating on microscopic-organism encounters, sprites, and level layouts directly in the browser.

WHAT IT DOES

- Starts random sessions or loads bundled and custom JSON levels from disk.
- Simulates the player, particles, enemies, eggs, macrophages, stentors, stones, obstacles, and oldbodies.
- Runs per-frame updates for motion, spawning, absorption, hatching, bounds control, and collision resolution.
- Renders the world, background image, and optional debug overlays to an HTML canvas.
- Provides live tuning controls for world parameters, sprite themes, and per-instance values while a session is running.
- Includes a level editor for placing entities, editing properties and presets, calibrating sprites, and saving level JSON.

HOW IT WORKS

- UI layer: index.html contains the launcher and in-game drawer; LEVEL_DESIGN/final_editor/index.html contains the separate editor UI.
- Runtime core: world.js re-exports world/init.js, world/loadLevel.js, world/update.js, world/draw.js, and world/resize.js.
- Shared state: worldState.js stores entity arrays plus vectorField, spatialGrid, background, and debug flags.
- Data flow: launcher or editor loads JSON -> initWorldFromLevel() maps coordinates and instantiates entity classes -> updateWorld() advances entities and resolves collisions -> drawWorld() paints the canvas.
- Content pipeline: sprites and preset JSON live under src/; bundled starter levels are level.json and introlevel.json; exported levels are stored in LEVELS/.
- Backend/services: Not found in repo.

HOW TO RUN

- Serve the repo root over local HTTP. Repo evidence: index.html uses ES modules and fetches ./level.json and ./introlevel.json. Exact server command: Not found in repo.
- Open index.html in the browser-served root, then choose Play Now, Random Game, or Import JSON.
- Open LEVEL_DESIGN/final_editor/index.html for level authoring. Folder access support is surfaced in editor code for Chrome or Edge; exact browser requirement wording for gameplay: Not found in repo.

